

Florentin Coeurdoux

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RESEARCH INTERESTS

Generative models (flows, stochastic interpolants, and energy-based models), Bayesian inference, and optimal transport. Building principled methods that connect rigorous theory with large-scale real-world deployment.

WORK EXPERIENCE

Capital Fund Management (CFM)

Paris, France

VP, Quantitative Research

Jan 2024 – Present

- Lead a team of quantitative researchers building and deploying ML models for live alpha generation on futures markets, covering commodities, equity indices, FX, bonds, short-term interest rates, and volatility products.
- Own the full pipeline from signal research to production deployment at one of the world's leading systematic hedge funds.

Research Scientist, CFM AI Lab

Jan 2024 – Present

- Conduct research on generative models and statistical inference, spanning flows, stochastic interpolants, and energy-based models, in close collaboration with E. Vanden-Eijnden (Courant/NYU) and S. Mallat (ENS/Flatiron Institute).
- Published 6 papers in 2025–2026, including at NeurIPS 2025 and ICML 2026; co-investigated applications of random matrix theory to ML with J.-P. Bouchaud (CFM).
- Supervise PhD students and research interns.

University of Oxford, Department of Statistics

Oxford, UK

Visiting Research Scholar

Oct 2022 – Mar 2023

- Conducted research on Monte Carlo methods and deep generative models under the supervision of Prof. Arnaud Doucet.
- Investigated connections between Langevin dynamics, score-based models, and normalizing flows for scalable posterior inference; contributed to work published in the *Machine Learning* journal.

AssessFirst

Paris, France

Head of Data Science

Dec 2021 – Nov 2022

- Led all data science initiatives and set the AI/ML roadmap for the company.
- Developed a variational inference algorithm served to 100k users, accelerating inference by 30%.
- Designed a statistical pattern detection algorithm for hiring, securing a 1.1M€ contract.

Credit Mutuel Arkéa

Rennes, France

Quantitative Developer

Apr 2019 – Nov 2020

- Developed a multivariate time series forecasting model for overdraft risk, deployed in production for 1.2 million users.
- Designed a deep reinforcement learning agent for dynamic budget allocation, enabling personalised financial recommendations at scale.
- Migrated option pricing codebase from VBA to C++, reducing execution time by 25% via importance sampling.

EDUCATION

Institut National Polytechnique (INP) – ANITI

Toulouse, France

Ph.D. in Applied Mathematics and Statistics

Nov 2020 – Dec 2023

- Supervised by Prof. Nicolas Dobigeon (IRIT/INP) and Prof. Pierre Chainais (Centrale Lille). Thesis on Monte Carlo sampling and deep generative models for Bayesian inference.

PUBLICATIONS

- 📄 E. Moreau, **F. Coeurdoux**, G. Ferré, E. Vanden-Eijnden, “Probing the Geometry of Diffusion Models with the String Method”, *ICML*, 2026. [arXiv]
- 📄 **F. Coeurdoux**, E. Lempereur, N. Cuvelles-Magar, S. Mallat, E. Vanden-Eijnden, “Generative Modeling via Kernelized Stochastic Interpolants”, *ICML Workshop*, 2026. [arXiv]
- 📄 E. Lempereur, N. Cuvelles-Magar, **F. Coeurdoux**, S. Mallat, E. Vanden-Eijnden, “MGD: Moment Guided Diffusion for Maximum Entropy Generation”, *MFML Workshop*, 2026. [arXiv]
- 📄 M. Chalvidal, **F. Coeurdoux**, E. Vanden-Eijnden, “Covariance Shrinkage via Stochastic Interpolation”, *Preprint*, 2026. [arXiv]
- 📄 **F. Coeurdoux**, G. Ferré, J.-P. Bouchaud, “Random Matrix Theory of Early-Stopped Gradient Flow”, *Preprint*, 2026. [arXiv]
- 📄 H. Negrel, **F. Coeurdoux**, M. Albergo, E. Vanden-Eijnden, “Multitask Learning with Stochastic Interpolants”, *NeurIPS*, 2025. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Plug-and-Play Split Gibbs Sampler: Embedding Deep Generative Priors in Bayesian Inference”, *IEEE Transactions on Image Processing*, 2024. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Normalizing Flow Sampling with Langevin Dynamics in the Latent Space”, *Machine Learning*, 2023. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Learning Optimal Transport Between Two Empirical Distributions with Normalizing Flows”, *ECML-PKDD*, 2022. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Sliced-Wasserstein Normalizing Flows: Beyond Maximum Likelihood Training”, *ESANN*, 2022. [arXiv]

SERVICE

Reviewer: NeurIPS, ICML, ICLR, AISTATS.

SKILLS

Programming: Python, C++.

ML/DL: PyTorch, JAX, CUDA.

Infrastructure: Distributed Training, Docker, Git, Linux, AWS.

Languages: French (Native), English (Fluent).

REFERENCES

Available upon request.